SHARP EQUILIBRIUM COUNTS WITH CUBIC SOURCES

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Abstract. A mass action network with n species, n reactions and stoichiometric rank $n-1$ has at most n positive nondegenerate equilibria on a stoichiometric class. We construct a family attaining this bound for every $n \geq 2$, with positive rational rate constants and maximum source molecularity three. The construction answers the sharpness question of Banaji and Feliu; their quadratic obstruction then proves that source molecularity three is minimal. The equilibrium set is an explicit monomial curve. Sparse polynomial interpolation gives exactly n simple intersections with one class, and transversality proves nondegeneracy. The count persists under small changes of rates and class level. The classes are unbounded, product molecularities may exceed three, and simultaneous stability is not asserted.

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1. The sharpness question

An (n, m, r) mass action network has n species, m reactions and stoichiometric rank r. Banaji and Feliu prove that every $(n, n, n-1)$ network has at most n positive nondegenerate equilibria on a stoichiometric class [2, Corollary 4.21(i)]. Their Remark 4.23 asks whether the bound is achieved for general n and, if so, what minimum source molecularity is needed. We answer both questions with a family whose sources have molecularity at most three. Attainment for $n = 2$ is already established in their Remark 4.23; the contribution is a uniform construction for all $n \geq 3$ and the resulting sharp source-molecularity threshold.

The monomial parametrisation of equilibria and the upper bound are part of the existing theory. The construction below combines a fixed cubic source matrix, sparse interpolation, and a choice of valid reaction products to attain the bound. The source degree stays three as n grows, although the degree of the scalar polynomial used in the construction does not stay bounded.

An anonymous, unrefereed manuscript [1] presents a quadratic $(n, n+2, n)$ family attaining $n+1$ positive nondegenerate equilibria, addressing the full-rank question in Banaji and Feliu's Remark 5.8. The present construction addresses their separate one-conservation-law question in Remark 4.23 and its minimum source molecularity.

Write reaction i as $y_i \to y_i + \gamma_i$, with nonnegative integral source and product vectors. Let A have rows $y_i^\top$ and Γ have columns γ_i. The mass action vector field is

$$\dot{x} = \Gamma v(x), \qquad v_i(x) = \kappa_i x^{y_i}, \qquad x \in \mathbb{R}^n_{>0}. \tag{1}$$

Its positive stoichiometric class through x is $(x + \operatorname{im}\Gamma) \cap \mathbb{R}^n_{>0}$. An equilibrium is nondegenerate when the Jacobian restricted to $\operatorname{im}\Gamma$ is invertible. The full Jacobian is singular when a conservation law is present.

Theorem 1. *For every integer $n \geq 2$, there exists an $(n, n, n-1)$ mass action network with positive rational rate constants and exactly n positive equilibria on one stoichiometric class, all nondegenerate. Its maximum source molecularity is three, the least possible for attaining this count. Its product molecularities are at most $2n$. The class is unbounded and its conservation vector has both signs.*

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2. Cubic sources and a sparse interpolation polynomial

Let $e_1, \dots, e_n$ be the standard basis, and fix the sources

$$y_i = 2e_i + e_{i+1} \quad (1 \le i < n), \qquad y_n = 2e_n. \tag{2}$$

The source matrix A is upper bidiagonal with diagonal entries two and superdiagonal entries one, so $\det A = 2^n$. Put

$$M = 2^n, \qquad p_i = \frac{2^n + (-1)^{n-i} 2^{i-1}}{3} \quad (1 \le i \le n). \tag{3}$$

These are distinct positive integers: the two terms in the numerator cancel modulo three, its value is positive, and the signed powers $(-1)^{n-i}2^{i-1}$ are distinct. They satisfy

$$2p_i + p_{i+1} = M \ (i < n), \qquad 2p_n = M, \qquad Ap = M\mathbf{1}. \tag{4}$$

Fix distinct positive rational numbers $r_1 < \cdots < r_n$; taking $r_j = j$ gives a concrete family for every n. Form the rational matrix

$$B_{j0} = 1, \qquad B_{ji} = r_j^{p_i} \quad (1 \le i \le n). \tag{5}$$

Choose a primitive integer vector $\alpha = (\alpha_0, \dots, \alpha_n)$ in its kernel, with the coefficient of the largest exponent positive. The next lemma shows that this choice is unique. It can be computed explicitly from the signed maximal minors of B, followed by division by their greatest common divisor.

Lemma 1. *The polynomial*

$$P(t) = \alpha_0 + \sum_{i=1}^{n} \alpha_i t^{p_i} \tag{6}$$

has all $n+1$ coefficients nonzero. Its positive roots are exactly $r_1, \dots, r_n$, each simple. The coefficients alternate in sign when ordered by increasing exponent.

Proof. Any n columns of B are independent. Otherwise their dependence would define a nonzero polynomial with at most n monomials and with n distinct positive roots, contrary to Descartes' rule of signs. Thus B has rank n and its one-dimensional kernel contains no vector with a zero coordinate. The polynomial (6) has $n+1$ nonzero monomials and already has n distinct positive roots. Descartes' rule bounds its positive roots, counted with multiplicity, by n. There are therefore no further positive roots, each prescribed root is simple, and all n possible coefficient sign changes occur. □

Define

$$c_i = |\alpha_i| > 0, \qquad z_i = \operatorname{sign}(\alpha_i), \qquad \rho_i = z_i z_{i+1} \in \{-1, 1\} \quad (i < n). \tag{7}$$

The n nonconstant coefficients alternate in their exponent order, so z has both signs for $n \ge 2$.

3. Valid products and the complete equilibrium set

Choose the stoichiometric reaction vectors

$$\gamma_i = -e_i + \rho_i e_{i+1} \quad (i < n), \qquad \gamma_n = -\sum_{i=1}^{n-1} \gamma_i. \tag{8}$$

For $i < n$ the product is $e_i + (1 + \rho_i)e_{i+1}$, a nonnegative integral vector of molecularity one or three. The final product is

$$y_n + \gamma_n = e_1 + \sum_{j=2}^{n-1} (1 - \rho_{j-1}) e_j + (2 - \rho_{n-1}) e_n. \tag{9}$$

It is also nonnegative and integral, with molecularity at most $2n$. All source and product complexes are therefore valid, and the maximum source molecularity is three. The sources are

pairwise distinct and every γ_i is nonzero: its ith entry is -1 for $i < n$, while the first entry of γ_n is 1. Thus there are exactly n distinct, nontrivial reactions.

The first $n-1$ columns of Γ are independent, as their first $n-1$ rows form a triangular matrix with diagonal -1. By construction $\Gamma\mathbf{1} = 0$ and $z^\top\Gamma = 0$. Hence

$$\operatorname{rank}\Gamma = n-1, \qquad \ker\Gamma = \mathbb{R}\mathbf{1}, \qquad \operatorname{im}\Gamma = \ker z^\top. \tag{10}$$

Choose the positive rational rate constants

$$\kappa_i = c^{-y_i} = \prod_{j=1}^{n} c_j^{-A_{ij}}. \tag{11}$$

At a positive equilibrium, (10) implies $v(x) = \lambda\mathbf{1}$ for some $\lambda > 0$. Taking componentwise logarithms gives

$$A\log(x/c) = (\log\lambda)\mathbf{1}.$$

Since $A^{-1}\mathbf{1} = p/M$, this is equivalent to

$$x_i(t) = c_i t^{p_i} \quad (1 \le i \le n), \qquad t = \lambda^{1/M} > 0. \tag{12}$$

Conversely, every point on (12) has all reaction fluxes equal to t^M and is an equilibrium. This is the full positive equilibrium set, not merely a collection of selected solutions.

Take the nonempty positive stoichiometric class

$$\mathcal{S} = \{x \in \mathbb{R}^n_{>0} : z^\top x = -\alpha_0\}. \tag{13}$$

It contains $x(r_1)$, and (10) shows that it is exactly one class. Along the equilibrium curve,

$$z^\top x(t) + \alpha_0 = P(t). \tag{14}$$

Lemma 1 therefore gives precisely the n positive equilibria $x(r_1), \dots, x(r_n)$ on $\mathcal{S}$.

4. Nondegeneracy and minimal source molecularity

At $x(t)$ the Jacobian of (1) is

$$J(t) = t^M \Gamma A \operatorname{diag}(x_1(t)^{-1}, \dots, x_n(t)^{-1}). \tag{15}$$

The factors to the right of Γ are invertible, so this matrix has rank $n-1$. Differentiating $\Gamma v(x(t)) = 0$ gives $J(t)x'(t) = 0$. Since $x'(t) \neq 0$, its kernel is exactly $\mathbb{R}x'(t)$. At each prescribed root,

$$z^\top x'(r_j) = P'(r_j) \neq 0. \tag{16}$$

Thus the kernel of $J(r_j)$ meets $\ker z^\top$ only at zero. The image of $J(r_j)$ lies in that same $(n-1)$-dimensional space, so its restriction is invertible. All n equilibria are nondegenerate.

Because z has both signs, there exists $w > 0$ with $z^\top w = 0$. For example, give each positive-sign coordinate the number of negative signs, and each negative-sign coordinate the number of positive signs. Then $x(r_1) + sw \in \mathcal{S}$ for all $s \ge 0$, proving unboundedness.

The construction has maximum source molecularity three. Conversely, [2, Corollary 5.16] proves that quadratic $(n, n, n-1)$ networks have at most two positive nondegenerate equilibria on a class, which rules out the count n for $n \ge 3$. The same corollary explicitly states that the smallest n for which two are attained by a quadratic network is three, ruling out attainment for $n = 2$ as well. Therefore no network in the specified class with maximum source molecularity at most two attains n. This proves the minimality assertion and completes Theorem 1.

Persistence under parameter perturbations. The count is not confined to the chosen rational rate vector. Keep the reaction network fixed and let $\kappa > 0$ vary. Its full positive equilibrium curve is still $x_i(t) = c_i(\kappa)t^{p_i}$, where

$$c(\kappa) = \exp(-A^{-1}\log\kappa).$$

On the class $z^\top x = K$, the scalar equation is $\sum_i z_i c_i(\kappa) t^{p_i} - K = 0$. The implicit function theorem preserves the n simple positive roots for all (κ, K) sufficiently close to the constructed

pair. This polynomial has at most $n+1$ monomials, so Descartes' rule excludes any additional positive root, including a degenerate one. Thus exactly n positive nondegenerate equilibria persist on an open neighbourhood in rate-constant and class-level space. No quantitative neighbourhood size or dynamical stability assertion is needed.

5. A three-species example and the scope of the result

For $n=3$, the powers are $(p_1,p_2,p_3)=(3,2,4)$. Choosing roots $1,2,3$ yields

$$P(t)=11t^4-60t^3+85t^2-36=(t-1)(t-2)(t-3)(11t+6). \tag{17}$$

Thus $c=(60,85,11)$ and $z=(-1,1,1)$. The reactions and rates are

$$\begin{aligned} 2X+Y &\xrightarrow{1/306000} X,\\ 2Y+Z &\xrightarrow{1/79475} Y+2Z,\\ 2Z &\xrightarrow{1/121} X+2Y+Z. \end{aligned} \tag{18}$$

The full positive equilibrium curve is $(X,Y,Z)=(60t^3,85t^2,11t^4)$, with common reaction flux t^8. The class $-X+Y+Z=36$ has exactly the three equilibria below.

t	X	Y	Z	$P'(t)$
1	60	85	11	34
2	480	340	176	-28
3	1620	765	891	78

The final column certifies transversality and nondegeneracy, not stability. The last product has molecularity four, illustrating why a bound on source molecularity is different from a bound on every reaction complex.

The conserved linear quantity has mixed signs, so it is not a conserved positive total mass. Our family is consistent with the upper bound two for bounded classes in [2, Corollary 4.21(ii)]; for $n>2$, unboundedness is necessary to attain n. The result counts equilibria and does not assert n stable states, a closed mass-balanced chemical mechanism, or an experimental implementation.

As separate finite checks, run from this directory:

```
python3 -B verification/verify_cubic_sharpness.py
  python3 -B -O verification/independent_cubic_check.py
```

The first reconstructs the sources, products, rational rates and restricted Jacobians for $2\le n\le 8$, checking all 35 prescribed equilibria with exact integer and rational arithmetic. The second independently differentiates the mass action equations using only standard-library rational arithmetic. Results are recorded in `verification/cubic-sharpness-certificate.json`. These checks corroborate the formulas; the all-n theorem rests on the proof above.

Research assistance

Valyu DeepResearch supported literature retrieval and source comparison. Coding agents developed the construction and analytic argument, implemented the exact verification programs, and drafted and revised the manuscript. The supplied finite checks corroborate the formulas; they do not replace the all-n proof or constitute external peer review.

Contributions and AI use

Harvey Yorke directed the project. Coding agents developed the arguments, the verification checkers and the manuscript drafts. Valyu DeepResearch carried out the literature, priority and internal adversarial reviews. These reviews are internal; the manuscript has not undergone external peer review. Harvey Yorke has not yet completed a line-by-line review of every proof;

correctness rests on the written proofs and on the checks and any formal verification described in this paper.

References

1. Anonymous, *Sharp positive-equilibrium counts in quadratic mass-action networks with two more reactions than species*, Publication candidate, version 2, 4 October 2026, 2026, Zenodo record 23146115.
2. Murad Banaji and Elisenda Feliu, *Positive equilibria in mass action networks: geometry and bounds*, Journal of Mathematical Biology **93** (2026), 37, doi:10.1007/s00285-026-02429-8.